WORKED SOLUTION

STEP I 2009, Question 4

Pure • Cosine Rule

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s1-q4

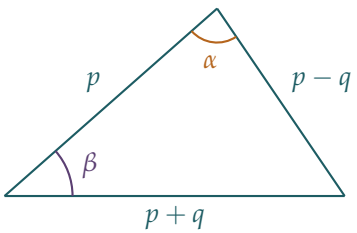
The key idea

In a triangle the largest angle is opposite the longest side and the smallest angle is opposite the shortest side. So the cosine rule gives $\cos\alpha$ and $\cos\beta$ directly in terms of p and q, and the identity is just algebra. For the second part, $\alpha = 2\beta$ lets us write $\cos\alpha$ in terms of $\cos\beta$ with a double-angle formula. The identity then becomes an equation in $\cos\beta$ alone.

Solution

The identity

Since $p - q < p < p + q$, the largest angle α is opposite the side $p + q$ and the smallest angle β is opposite the side $p - q$.



The cosine rule for α gives

$$\cos\alpha = \frac{p^2 + (p-q)^2 - (p+q)^2}{2p(p-q)} = \frac{p^2 - 4pq}{2p(p-q)} = \frac{p-4q}{2(p-q)},$$

and for β it gives

$$\cos\beta = \frac{p^2 + (p+q)^2 - (p-q)^2}{2p(p+q)} = \frac{p^2 + 4pq}{2p(p+q)} = \frac{p+4q}{2(p+q)}.$$

We cancelled p, which is allowed because $p > 0$. Then

$$1 - \cos\alpha = \frac{2(p-q) - (p-4q)}{2(p-q)} = \frac{p+2q}{2(p-q)},$$
$$1 - \cos\beta = \frac{2(p+q) - (p+4q)}{2(p+q)} = \frac{p-2q}{2(p+q)},$$

so

$$4(1-\cos\alpha)(1-\cos\beta) = \frac{(p+2q)(p-2q)}{(p-q)(p+q)} = \frac{p^2-4q^2}{p^2-q^2}.$$

On the other side,

$$\begin{aligned}\cos\alpha + \cos\beta &= \frac{(p-4q)(p+q) + (p+4q)(p-q)}{2(p^2-q^2)} \\ &= \frac{(p^2-3pq-4q^2) + (p^2+3pq-4q^2)}{2(p^2-q^2)} = \frac{p^2-4q^2}{p^2-q^2}.\end{aligned}$$

The two expressions agree, so $4(1-\cos\alpha)(1-\cos\beta) = \cos\alpha + \cos\beta$, as required.

(For the triangle to exist we need $(p-q) + p > p + q$, that is, $p > 2q$. Then both sides of the identity are positive, but the algebra above doesn't depend on this.)

The case $\alpha = 2\beta$

Write $c = \cos\beta$. Then $\cos\alpha = \cos 2\beta = 2c^2 - 1$, so $1 - \cos\alpha = 2(1-c^2) = 2(1-c)(1+c)$. Substituting into the identity gives

$$8(1-c)^2(1+c) = 2c^2 + c - 1 = (2c-1)(c+1).$$

Ask yourself Can we divide both sides by $1+c$?

Yes. β is an angle of a triangle, so $0 < \beta < \pi$ and $c = \cos\beta > -1$, which means $1 + c \neq 0$. Dividing by it gives

$$8(1-c)^2 = 2c - 1 \quad\Longrightarrow\quad 8c^2 - 18c + 9 = 0 \quad\Longrightarrow\quad (4c-3)(2c-3) = 0.$$

So $c = \frac{3}{4}$ or $c = \frac{3}{2}$. A cosine can't exceed 1, so we reject $\frac{3}{2}$. Therefore $\cos\beta = \frac{3}{4}$, as required.

The ratio of the sides

Using the formula for $\cos\beta$ from the first part,

$$\frac{p+4q}{2(p+q)} = \frac{3}{4} \quad\Longrightarrow\quad 4p + 16q = 6p + 6q \quad\Longrightarrow\quad p = 5q.$$

This satisfies $p > 2q$, so the triangle exists. Its sides are $p - q = 4q$, $p = 5q$ and $p + q = 6q$, so the ratio of the sides is

$$4 : 5 : 6.$$

Checks

In the $4,5,6$ triangle, the cosine rule gives $\cos\beta = \frac{25+36-16}{60} = \frac{3}{4}$ and $\cos\alpha = \frac{16+25-36}{40} = \frac{1}{8}$. Also $2\cdot(\frac{3}{4})^2 - 1 = \frac{1}{8}$, which confirms that $\alpha = 2\beta$ (both angles lie in $(0,\pi)$, where cosine is one-to-one). Numerically, $\beta \approx 0.7227$ and $\alpha \approx 1.4455$ radians. In this triangle the identity reads $4\cdot\frac{7}{8}\cdot\frac{1}{4} = \frac{7}{8} = \frac{1}{8} + \frac{3}{4}$. We also tested the identity on 10^5 random triangles with $p > 2q$, and the two sides agreed to within 3×10^{-15}.

Common slips

- Matching the angles to the wrong sides. The largest angle is opposite the *longest* side, $p+q$.
- Cancelling the factor $1+c$ without saying why it is non-zero, or keeping the root $c = \frac{3}{2}$.
- Giving the answer as $p = 5q$ instead of the ratio of the three sides, $4:5:6$.

Beyond STEP

Up to scaling, the $4,5,6$ triangle is the only triangle whose sides are in arithmetic progression and whose largest angle is twice its smallest. Questions like "which triangles have integer sides and angles in a given ratio?" lead quickly to Diophantine equations and to parametrising rational points on curves, much as the Pythagorean triples do. Davenport's *The Higher Arithmetic* is a good place to see these ideas develop, and Siklos's *Advanced Problems in Mathematics* has more STEP-style geometry and trigonometry.

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